FROM THE EQUILATERAL TRIANGLE TO THE FLAT LIMIT: THE ZEE-TO-CALIPER FRONTIER IN BELLMAN'S FOREST PROBLEM

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Abstract. Let T_α be the isosceles triangle with unit legs and apex angle $60^\circ \leq \alpha < 180^\circ$. At the equilateral end the Besicovitch–Movshovich zee is optimal through $\alpha = 90^\circ$; at the flat end the natural construction is Zalgaller's strip caliper. We study the unresolved upper-bound frontier between them.

Our exact result removes Ward's proposed square phase. We construct an explicit piecewise-polynomial family of trapezoidal staples which escapes T_α and is shorter than every escaping square path for $95.6734^\circ \leq \alpha \leq 117.8562^\circ$. On $95.6734^\circ \leq \alpha \leq 117.0062^\circ$ it also beats Ward's other three comparators: the zee, the diameter, and the scaled caliper. Escape is reduced to containment of a disk in one Minkowski sum, and the resulting 116 polynomial inequalities are certified exactly by Sturm's theorem.

Numerical continuation supplies the global synthesis. The best paths found after the zee lie on one line–arc Zalgalloid branch: it crosses the zee transversally at $\alpha_\times \approx 95.43^\circ$ and meets the caliper tangentially near $\alpha^* \approx 124.8^\circ$, thereafter being the caliper. Thus the current candidate upper envelope from the equilateral triangle to the flat limit has a single nonsmooth transition, rather than distinct square, trapezoid, Zalgalloid, and caliper phases. We also give an exact line–arc anchor at $\alpha = 108^\circ$. Optimality beyond $\alpha = 90^\circ$ remains open.

1. Introduction

Bellman's lost-in-a-forest problem asks for the shortest path that reaches the boundary of a known forest when the starting position and heading are unknown [2, 5]. Equivalently, a rectifiable path P is an *escape path* for a compact convex region K if no congruent copy of P lies in $\operatorname{int} K$. We write $\ell(K)$ for the infimum of the lengths of such paths.

We study the one-sided isosceles family from the equilateral triangle to the flat limit. Thus T_α has unit equal legs and apex angle $60^\circ \leq \alpha < 180^\circ$. The complementary acute-apex family $0^\circ < \alpha < 60^\circ$ is a different problem and is not treated here. For the exact proofs we use the base angle

$$\beta = \frac{180^\circ - \alpha}{2}, \qquad 0^\circ < \beta \leq 60^\circ,$$

because $u = \tan(\beta/2)$ rationalizes the triangle geometry.

At the equilateral end the best understood construction is the Besicovitch–Movshovich *zee*: a three-segment zigzag that is optimal for $60^\circ \leq \alpha \leq 90^\circ$, equivalently base angles $45^\circ \leq \beta \leq 60^\circ$, [3, 4, 9, 8]. In the opposite, thin-triangle limit, the forest approaches a strip of width $\sin\beta$, and Zalgaller's strip solution gives the upper bound

$$\ell(T_\alpha) \leq \zeta \sin\beta, \qquad \zeta = 2.2782916\ldots$$

[11, 1]. What happens between these two mechanisms was not known.

Ward [10] compared the zee, the scaled caliper, the diameter, and a path along three sides of a square. Among those four candidates the square led for $32.36^\circ < \beta < 41.34^\circ$. He also reported

Date: July 22, 2026.

The computations, verification scripts, and initial draft were produced with the assistance of Claude Fable 5 (Anthropic). All exact certificates are supplied as ancillary files.

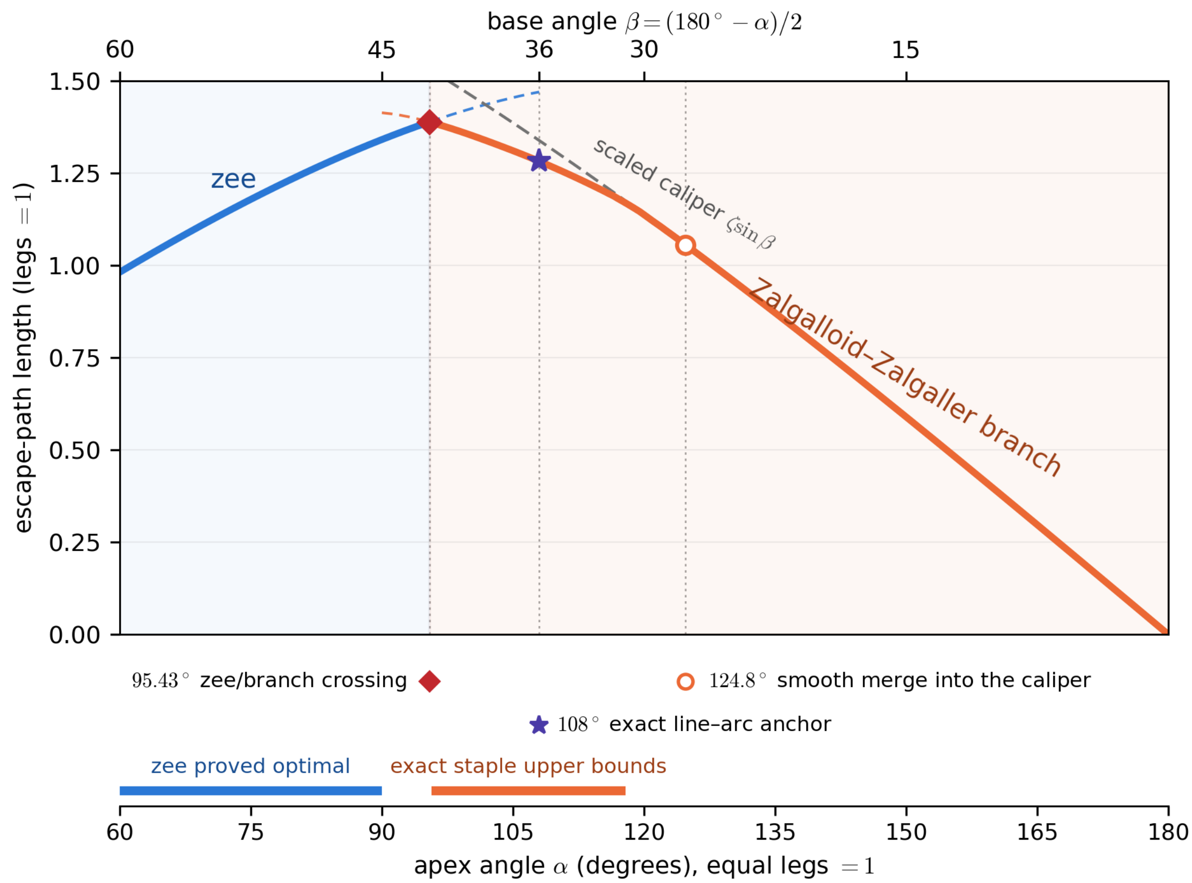

FIGURE 1. The current upper-bound frontier on the range treated in this paper. The zee is proved optimal through $\alpha = 90°$ and remains the best known candidate until the unique nonsmooth crossing $\alpha_\times \approx 95.43°$. The orange curve is one connected Zalgalloid–Zalgaller family: it has line–arc form in the middle and meets the caliper smoothly near $\alpha^* \approx 124.8°$, with no second phase transition. Thin dashed curves continue each family past the crossing and show the scaled caliper approaching the branch; the bars on the baseline mark rigorous status. The curves and transition locations are numerical outside Theorem 2.1 and Proposition 4.1.

that releasing the right angles to obtain an isosceles trapezoidal hull made the path shorter, and asked whether a family of "Zalgalloids" might interpolate between the new bounded-forest path and Zalgaller's caliper. Gibbs later found trapezoidal and line–arc candidates numerically [6].

The apparent proliferation of phases is misleading. Our contribution has three parts.

First, we turn Ward's trapezoidal observation into an exact interval theorem. A rational-polynomial family of three-segment *staples* beats every escaping square path on an interval strictly containing Ward's entire square range, and beats all four of his comparison families on a slightly smaller interval. Thus the square does not define a phase of the best known upper envelope.

Second, we give a finite geometric escape certificate for arbitrary convex hulls in a triangle. This converts the continuum of unknown positions and headings into support inequalities for a single Minkowski sum, making the staple theorem amenable to exact verification.

Third, continuation of the sharper line–arc paths shows that Ward's Zalgalloid is not an additional phase between a trapezoid and the caliper. It is one connected family: near the zee it resembles a slightly rounded trapezoid; as β decreases its arcs grow and its central bar shrinks; at the lower end the bar disappears and the path is Zalgaller's caliper. Consequently the computed upper envelope has only one change of family, where the zee crosses the Zalgalloid branch. Figure 1 is the whole story on the range treated here.

The logical distinction is important. Theorem 2.1 and Proposition 4.1 are exact upper bounds. The global continuation, the assertion that no still shorter family has been missed, and both event locations are numerical. We claim neither a classification nor optimality for $\alpha > 90°$; we identify and certify a substantially improved upper-bound frontier.

2. The exact bridge beyond the zee

Write T_α for the isosceles triangle with unit legs, and retain $\beta = (180° - \alpha)/2$. A *staple* is the three-segment path through

$$(-a, c), \quad (-b, 0), \quad (b, 0), \quad (a, c), \qquad a, b, c > 0.$$

Its convex hull is an isosceles trapezoid and its length is

$$|S| = 2b + 2\sqrt{(a-b)^2 + c^2}.$$

The three-sided square path is the special case $a = b$ and $c = 2b$.

Theorem 2.1 (certified staple interval)**.** *Let $u = \tan(\beta/2)$. There is an explicit three-piece family $S(u)$ whose parameters $a(u), b(u), c(u)$ are rational polynomials of degree at most 6, specified in the ancillary file `anc/interval_construction.json`, such that for every*

$$95.6734° \le \alpha \le 117.8562° \quad \bigl(31.0719° \le \beta \le 42.1633°\bigr),$$

the following hold:

(1) *$S(u)$ escapes T_α;*
(2) *$|S(u)| < 3s$ for every square path of side s that escapes T_α;*
(3) *$|S(u)|$ is shorter than the diameter and the zee throughout the interval, and shorter than the scaled caliper whenever $\alpha \le 117.0062°$ (equivalently $\beta \ge 31.4969°$).*

Corollary 2.2. *For every $95.68° \le \alpha \le 117.00°$, an escaping staple is strictly shorter than each of Ward's four candidates: every escaping square path, the diameter, the scaled Zalgaller caliper, and the zee. In particular, Ward's entire square-leading interval $97.32° < \alpha < 115.28°$ is covered by a strictly shorter certified path.*

The theorem removes the square from the candidate diagram, but the polygonal staple itself is not the final numerical answer. Rounding its two shoulders produces a slightly shorter escape path. The staple is therefore the exact, proof-friendly skeleton of the Zalgalloid branch, not a separate phase.

3. A finite escape certificate for triangles

For a compact convex set C, let $h_C(u) = \sup_{x \in C}\langle x, u\rangle$ be its support function, and let R_θ denote rotation through θ.

Lemma 3.1 (triangle fit criterion)**.** *Let* $T = \{x : \langle n_i, x\rangle \le t_i,\ i = 1, 2, 3\}$ *be a triangle with unit outward normals* n_i *and opposite side lengths* s_i*. A compact convex set* C *has a translate contained in* T *if and only if*

$$\sum_{i=1}^{3} s_i h_C(n_i) \le 2\,\mathrm{Area}(T).$$

Proof. The translation problem is the system $\langle n_i, x\rangle \le c_i := t_i - h_C(n_i)$. Since $\sum_i s_i n_i = 0$ and $\sum_i s_i t_i = 2\,\mathrm{Area}(T)$, feasibility implies $\sum_i s_i c_i \ge 0$, which is the displayed inequality. Conversely, put $\lambda = (\sum_i s_i c_i)/(2\,\mathrm{Area}(T)) \ge 0$. The three equations $\langle n_i, x_0\rangle = c_i - \lambda t_i$ are compatible because their only linear dependence has coefficients s_i. The feasible translation set is then the nonempty homothet $x_0 + \lambda T$. □

The lemma is the planar simplex case of a containment theorem of Lutwak [7]. For escape paths it gives the following finite geometric form.

Proposition 3.2 (Minkowski escape certificate)**.** *Let* $n_i = (\cos\alpha_i, \sin\alpha_i)$*, and let* $H = \mathrm{conv}(P)$*. The path* P *escapes* T *if and only if*

$$W_T(H) := \bigoplus_{i=1}^{3} s_i R_{-\alpha_i} H \supseteq \overline{D}\big(0, 2\,\mathrm{Area}(T)\big).$$

If H *is polygonal, the inclusion reduces to finitely many support inequalities, one for each edge-normal direction of the summands.*

Proof. A congruent copy of P fits in $\operatorname{int} T$ exactly when its convex hull does. By Lemma 3.1, P escapes precisely when, for every rotation R_θ,

$$\sum_i s_i h_{R_\theta H}(n_i) \ge 2\,\mathrm{Area}(T).$$

The left side is $h_{W_T(H)}(\cos(-\theta), \sin(-\theta))$. Requiring this support value in every direction is equivalent to the stated disk inclusion. □

This proposition is the proof engine for both the exact staple theorem and the line–arc checks. It eliminates all positions and headings; no sampling of placements occurs in an accepted certificate.

Proof of Theorem 2.1. The proof is computer-assisted but exact. Parametrize by $u = \tan(\beta/2)$, so

$$\sin\beta = \frac{2u}{1+u^2}, \qquad \cos\beta = \frac{1-u^2}{1+u^2}.$$

All triangle data in Proposition 3.2 are therefore rational functions of u. For a polynomial staple, every support inequality for the Minkowski sum becomes, after clearing positive denominators and squaring sign-controlled radicals, a polynomial inequality over $\mathbb{Q}$.

The construction has three polynomial pieces on

$$[0.278, 0.3530], \qquad [0.3530, 0.353399], \qquad [0.353399, 0.3855].$$

For each piece the twelve possible edge-normal directions of the Minkowski sum are assigned fixed support vertices. The resulting disk-containment conditions are lower bounds for the true support of $W_T(H)$ and hence prove escape.

To eliminate square paths, inscribe in T_α the two squares used by Ward. Their side lengths are

$$s_A = \frac{2\sin\beta\cos\beta}{\sin\beta + 2\cos\beta}, \qquad s_B = \frac{\sin\beta}{\sin\beta + \cos\beta}.$$

If a square of side s_0 lies in T_α, then every square path of side $s < s_0$ fits strictly inside T_α and cannot escape. It therefore suffices to prove $|S(u)| < 3s_A$ on one subinterval and $|S(u)| < 3s_B$ on the other. The diameter, zee, and caliper comparisons are handled in the same rational parametrization.

Altogether the ancillary verifier derives 116 rational polynomials of degree at most 40. On each rational interval it checks positive endpoint values and an exact Sturm root count of zero. The lower bound $\zeta > 2.2782916$ used for the caliper comparison is proved separately by rational root isolation and alternating Taylor bounds. All 116 certificates pass; floating point is used only to discover the family, not to accept a sign. □

We now sharpen that exact polygonal skeleton and follow it numerically to the thin-triangle limit.

4. One Zalgalloid–Zalgaller branch

The numerical optimizer replaces each shoulder of the staple by a circular arc and then continues the resulting line–arc path in β. Each path traverses the boundary of its convex hull except for the chord joining its two tips. The circular arcs have radius $\sin\beta$, the width of the triangle; this law is forced whenever the corresponding arc of the weighted Minkowski sum is tangent to the certificate disk.

At the golden gnomon, where $\alpha = 108°$ and $\beta = 36°$, the line–arc certificate can be made entirely exact.

Proposition 4.1 (exact line–arc anchor)**.** *For the golden gnomon there is a symmetric line–arc escape path with rational centers, radii, endpoints, and tangent lines whose length is*

$$2q + 2\sqrt{A} + 2\sqrt{B} + 2r\arccos C = 1.2826798604\ldots < 1.282680, \qquad q, A, B, C, r \in \mathbb{Q}.$$

It is shorter than the certified polygonal staple, whose length is $1.2849615334\ldots$. *It is also shorter than Ward's square path, whose length is* $1.293473869\ldots$.

Proof. The feature normals of the three rotated hulls partition the direction circle into eighteen windows. On each window one fixed vertex or circular arc supplies a lower bound for the support of $W_T(H)$. Endpoint signs and an antipode exclusion prove that the lower bound stays above the disk radius throughout the window. All algebraic signs are exact. Rational upper bounds for the square roots and a degree-12 alternating Taylor bound for cos give the displayed strict length inequality. The rational data and all eighteen checks are in `anc/arc_certificate.py`. □

Continuing this template toward increasing α produces the deformation in Figure 2. The central bar shortens and the shoulder arcs grow. The computed advantage over the caliper vanishes for $124.5° < \alpha < 125.0°$; by $\alpha = 125.5°$ the optimized bar has collapsed and the path agrees with the scaled Zalgaller caliper to solver precision. The length data are consistent with tangential contact near $\alpha^* \approx 124.8°$ ($\beta^* \approx 27.6°$). Thus no new construction takes over: numerically, the Zalgalloid branch simply becomes the caliper.

Continuing toward decreasing α gives a different event. The line–arc branch and the zee coexist and their lengths cross transversally at

$$\alpha_\times = 95.426° \quad (\beta_\times = 42.287°), \qquad L_\times = 1.38920.$$

The zee visits the vertices of its quadrilateral hull in diagonal order; the Zalgalloid follows the boundary of its hull. Neither continuation turns into the other. This is the only non-smooth change of family in the computed envelope.

We summarize the status precisely:

- the zee is proved optimal for $60° \le \alpha \le 90°$;

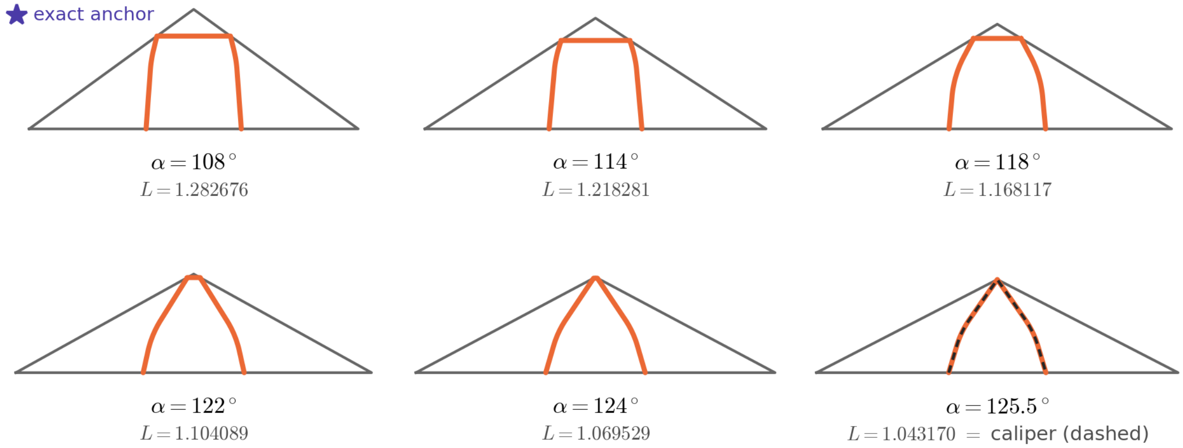

FIGURE 2. Numerical Zalgalloid continuation. As α increases (and β decreases), the bar shrinks and the arcs deepen; at $\alpha = 125.5°$ ($\beta = 27.25°$) the optimized member coincides with the scaled caliper (dashed). The rigorous anchors are Theorem 2.1 and Proposition 4.1.

- the staple theorem gives exact upper bounds and eliminates the square for $95.67° \leq \alpha \leq 117.86°$;
- the line–arc path at $\alpha = 108°$ is an exact Zalgalloid anchor;
- the global branch and both numerical events—the crossing at 95.43° and the merger into the caliper near 124.8°— remain numerical;
- optimality of the middle branch remains open.

5. Conclusion

On $60° \leq \alpha < 180°$, the candidate phase diagram for isosceles triangles is simpler than the historical list of constructions suggests. Ward's square is beaten throughout its range by a certified trapezoidal staple. Rounding the staple gives the sharper Zalgalloid, and continuing that path does not lead to a succession of new phases: it deforms continuously into Zalgaller's caliper. Among all families found, the only transition in the current candidate envelope is the crossing from the zee to this single Zalgalloid–Zalgaller branch.

What is proved here is the upper-bound part of that picture: an exact interval family eliminating the square, a finite Minkowski certificate, and an exact line–arc anchor. Proving the Zalgalloid branch optimal, or even certifying the full line–arc continuation uniformly in β, is the remaining problem.

Ancillary files. The arXiv source package includes four data and verifier files documented in `anc/README.txt`. They reconstruct every exact datum used above and verify the 116 interval certificates, the golden-gnomon polygonal certificate, and the eighteen line–arc windows. Discovery code and numerical continuation data are available at `https://github.com/atemerev/bellman-staple`.

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